

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1574

Hutchins 138 kV
November 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1574. The Interconnection Customer proposes a 320.5 MW wind facility interconnecting at the Hutchins 138 kV substation in the Virginia area. The request was received on September 24, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 552 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,751,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Geneva Border City to Keokuk 138 KV line	N Tower Tap to Dpc Elroy 138 KV	107.0	346	12.8
Nestuca to City West Diesel Plant 345 KV	Nauvoo to City W Diesel Plant 345 KV Line	102.8	1,471	5.0
Fredericksburg Substation to Polk City 345 KV Ckt 1	Fredericksburg to Polk City 345 KV	101.2	1,465	6.8
Fredericksburg to Polk City 345 KV #2	Fredericksburg to Polk City 345 KV #1	101.2	1,465	6.8
Danesville to Thomaston 345 KV line	Comanche Peak to Thomaston 345 KV	100.1	1,328	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Keokuk to Geneva Border City 138 KV	Reconductor line	\$1,541,000	23% (\$358,000)
Nestuca to City West Diesel Plant 345 KV	Reconductor line	\$9,744,000	9% (\$841,000)
Fredericksburg Substation to Polk City 345 KV Ckt 1	Reconductor line	\$27,228,000	3% (\$715,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fredericksburg to Polk City 345 KV #2	Reconductor line	\$46,350,000	3% (\$1,218,000)
Dranesville to Thomaston 345 KV	Reconductor line	\$16,095,000	4% (\$619,000)
	Total allocated to project		\$3,751,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.